TERRAFORM AUTOMATION PROJECT (JENKINS + SONARQUBE + TRIVY)

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Course: DevOps

Date: April 2026

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Project Overview

This project demonstrates Infrastructure as Code (IaC) using Terraform to automatically create an AWS EC2 instance and install DevOps tools.

The setup includes:

- Jenkins installation
- SonarQube using Docker
- Trivy security scanner
- Automated provisioning using Terraform

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Architecture / Workflow

1. Terraform creates EC2 instance
2. Security group opens required ports
3. User data script installs tools
4. Jenkins starts automatically
5. SonarQube runs in Docker container
6. Trivy is installed for scanning

Tools Used

- Terraform
- AWS EC2
- Jenkins
- Docker
- SonarQube
- Trivy

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Output Screenshots

- Terraform apply output
- EC2 instance created
- Jenkins running
- SonarQube UI (port 9000)
- Trivy installation

Conclusion

This project demonstrates Infrastructure as Code using Terraform. It automates server creation and tool installation, reducing manual setup time.

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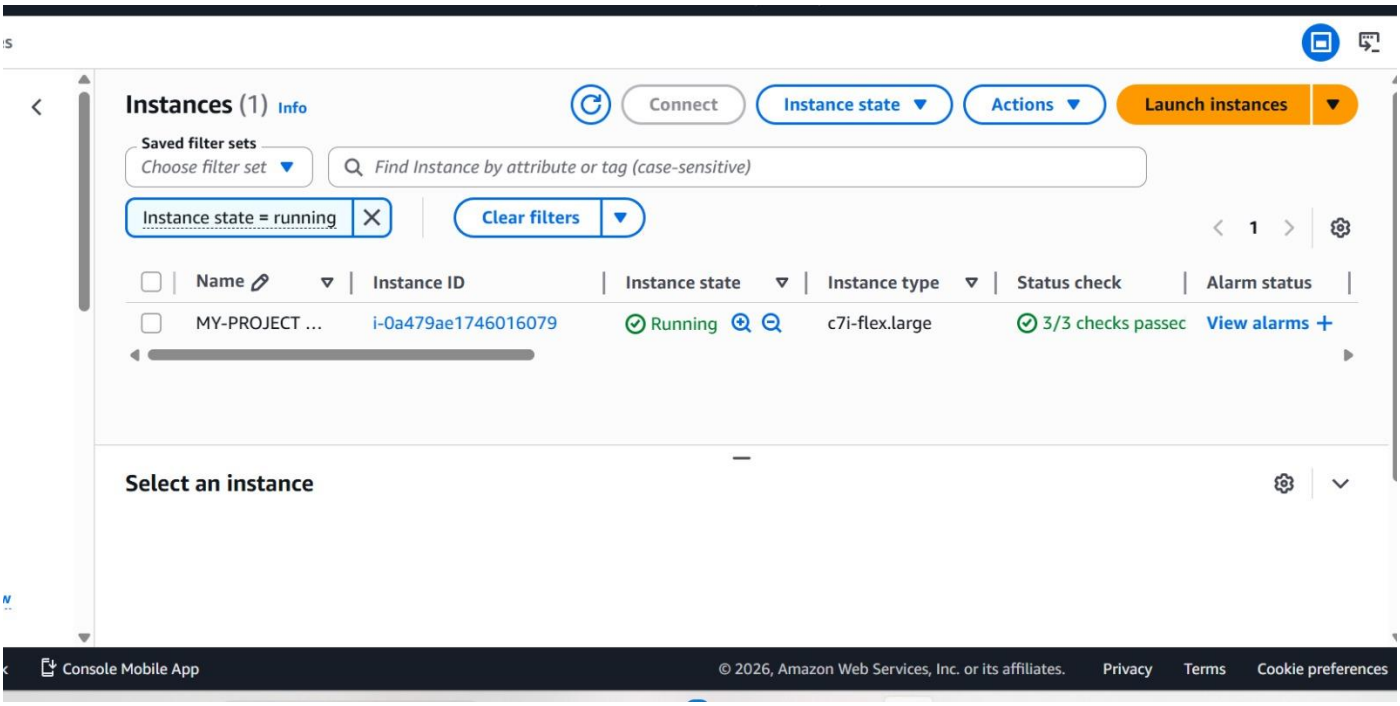

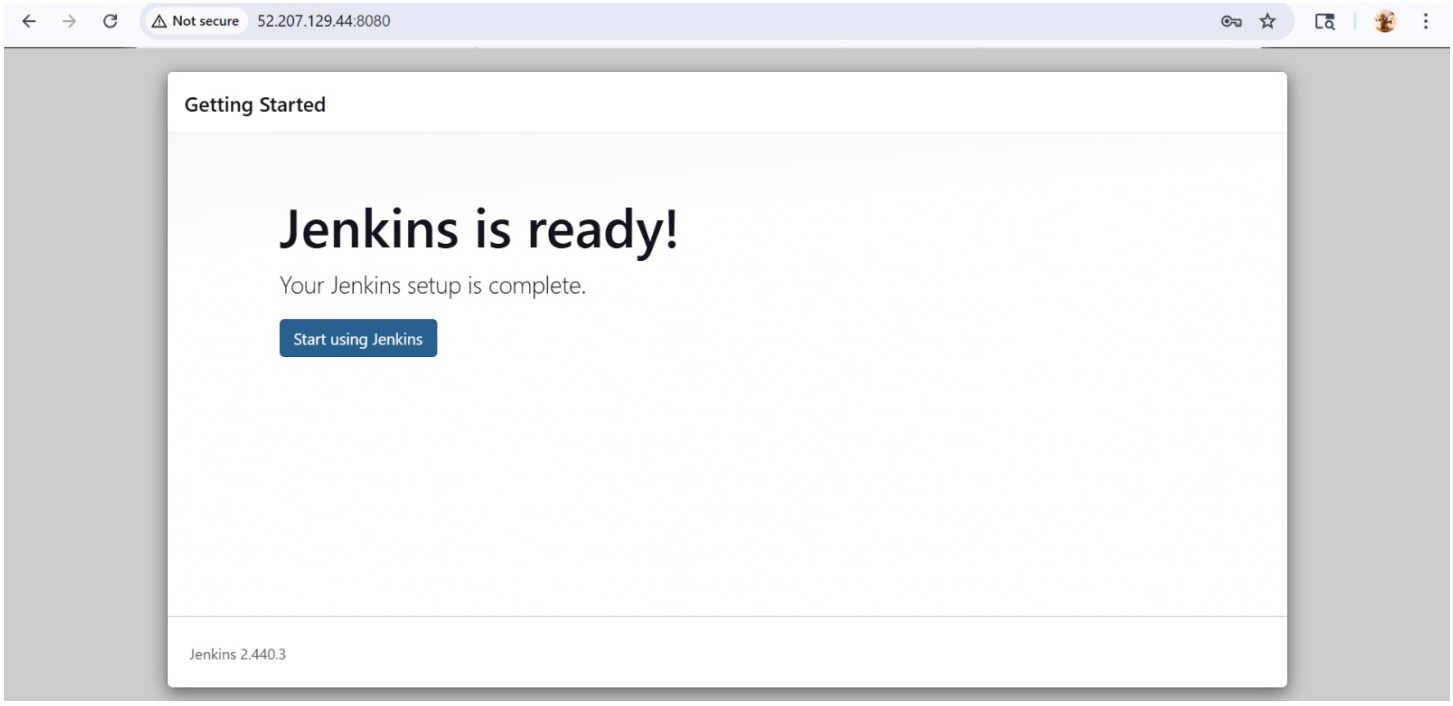
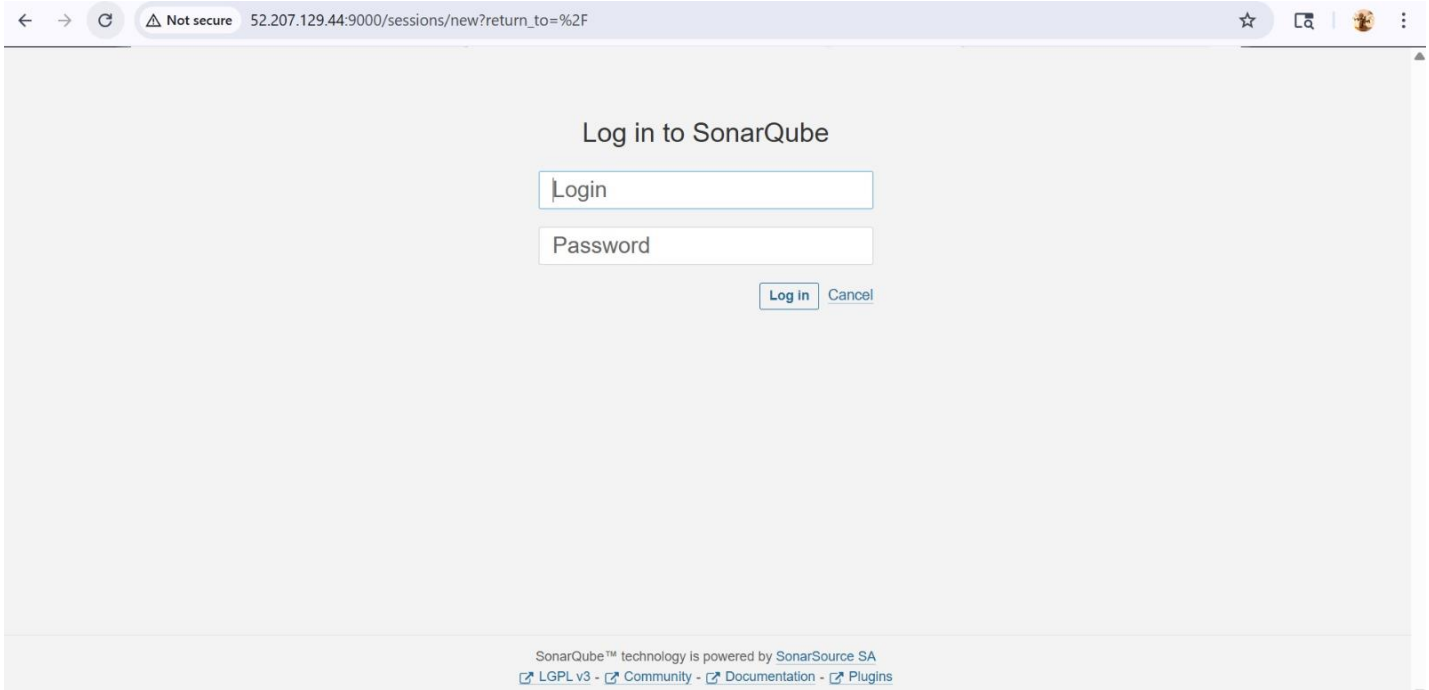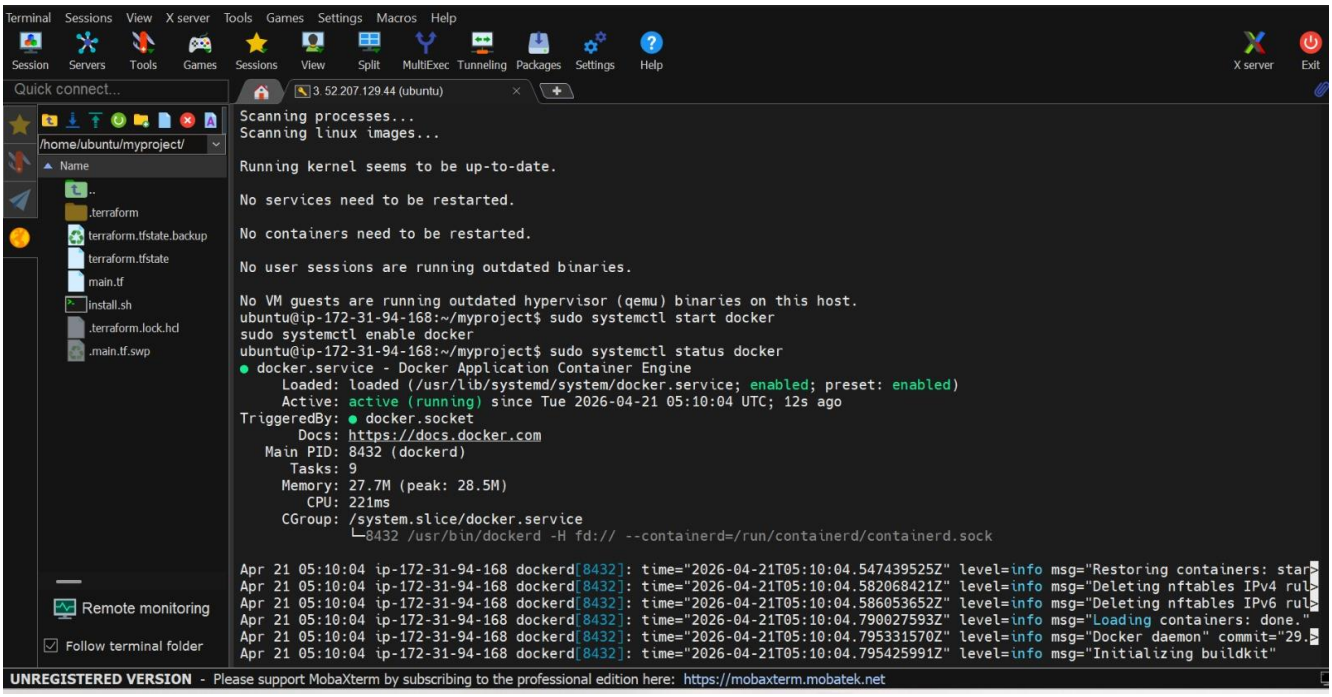
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